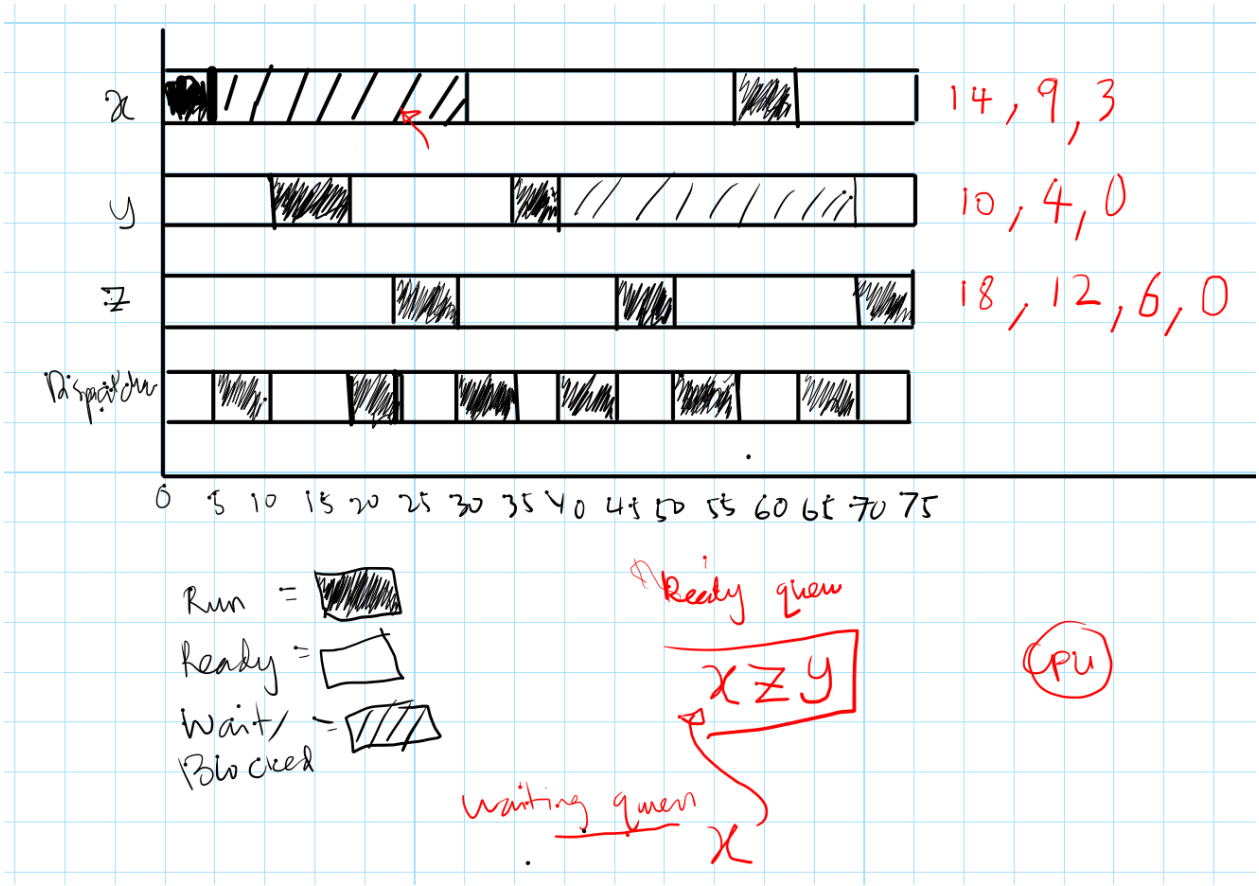
QUESTION 2

a) Show your working by using the gantt chart to trace the process states for up to 75 instruction cycles. You may assume that the three processes are run concurrently on a uniprocessor and that the maximum CPU time for each process is 6 instruction cycles.

[12 marks]



b) Which jobs has been **Unsuccessfully Executed**? [2 marks]

Process x

c) What is the **Turnaround Time** of **Process Z**? [2 marks]

Turnaround time = Completion time – Arrival time

75 – 0 = 75

d) What is the **Waiting Time** of **Process Z**? [2 marks]

Waiting time = Turnaround Time – Burst Time

75 – 18 = 57

e) What is **the states** of processes X, Y and Z at $\mathbf{t_{15}}$? [3 marks]

x = waiting/blocked

y = running

z = ready

f) What is **the states** of processes X, Y and Z at $\mathbf{t_{40}}$? [3 marks]

x = ready

y = waiting/blocked

z = ready